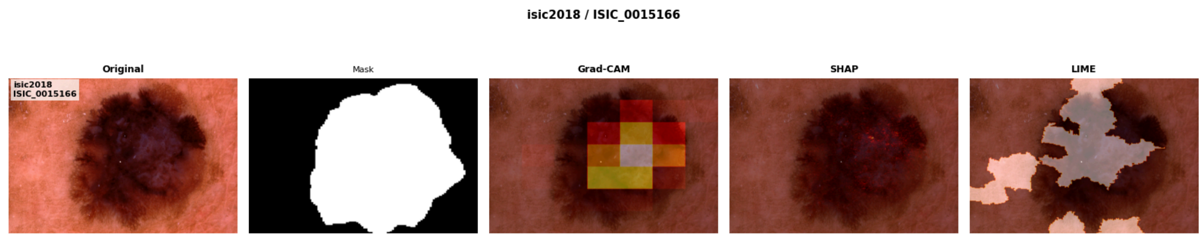
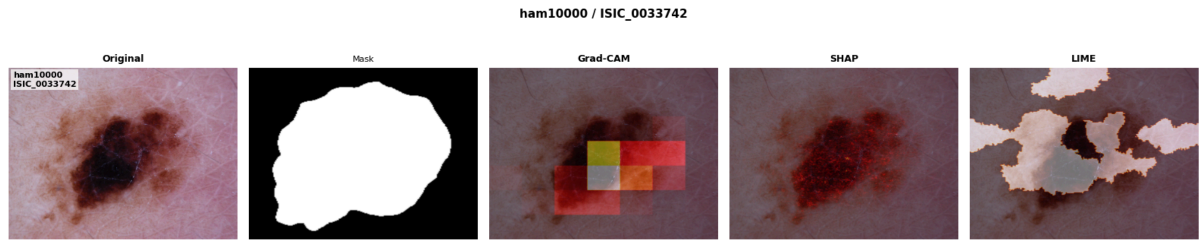
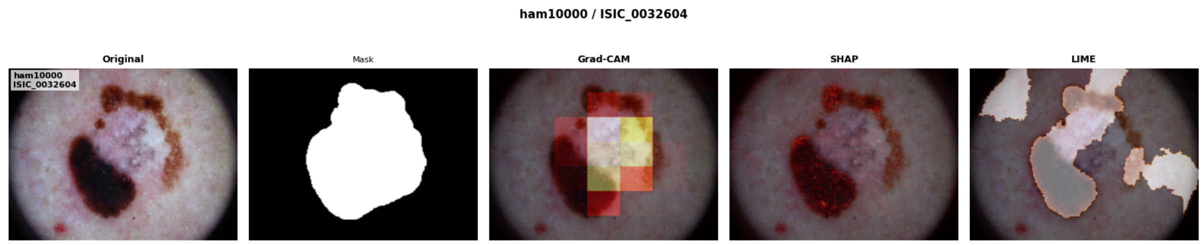
Appendix

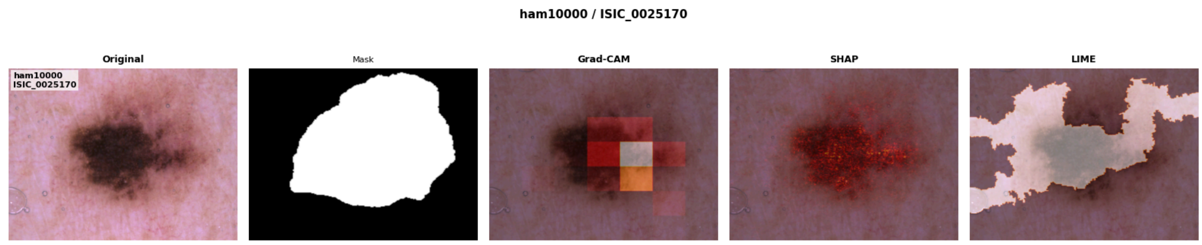
Section 1 - Extra images

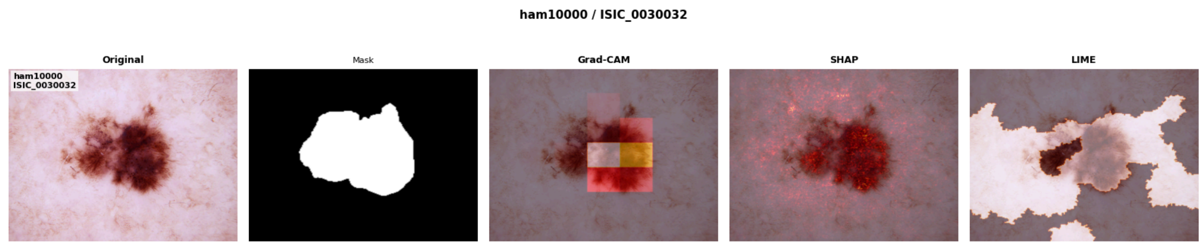
1.1 - XAI methods

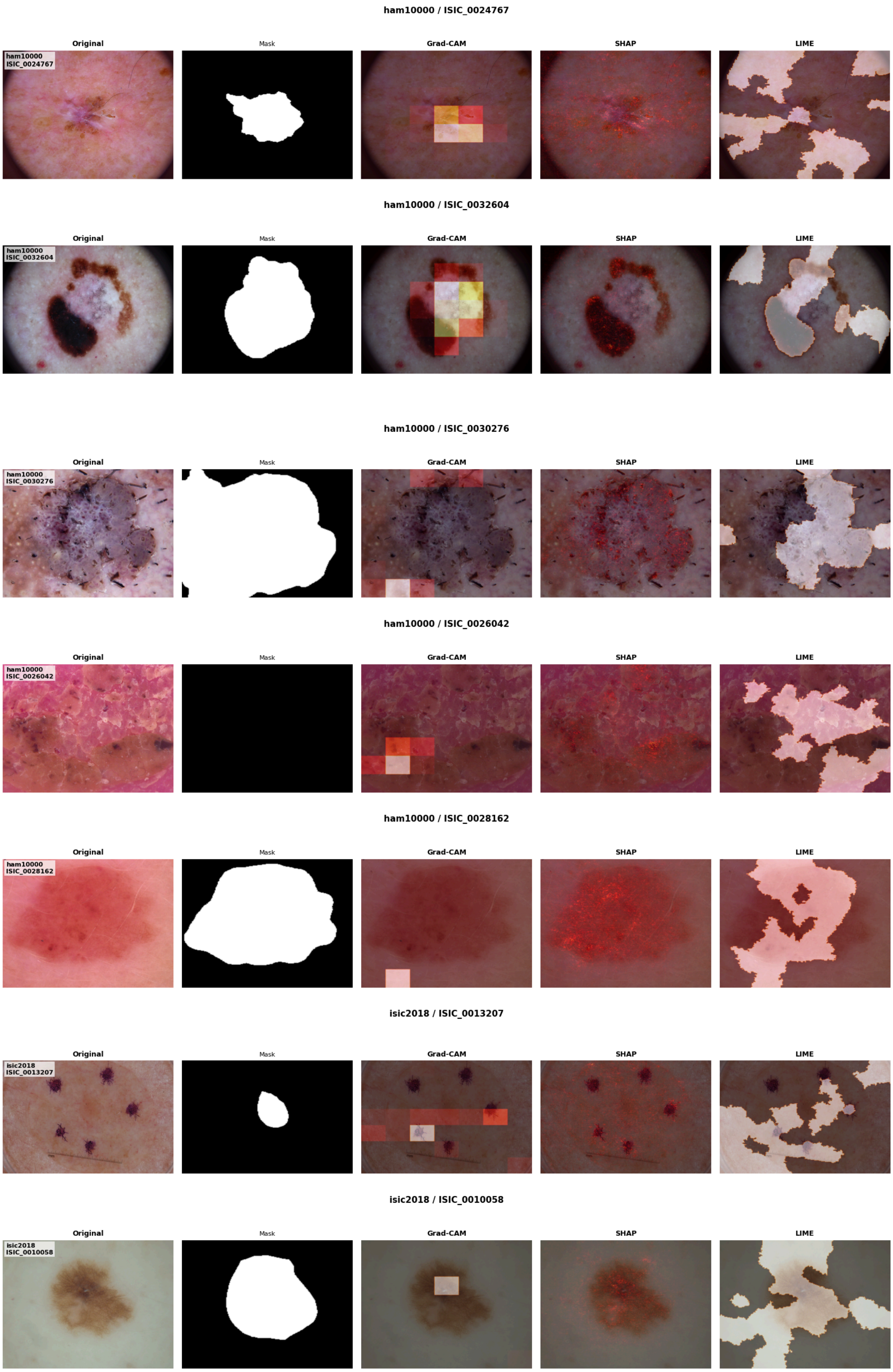




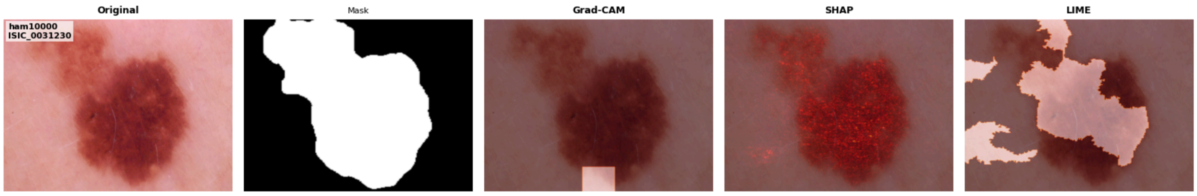






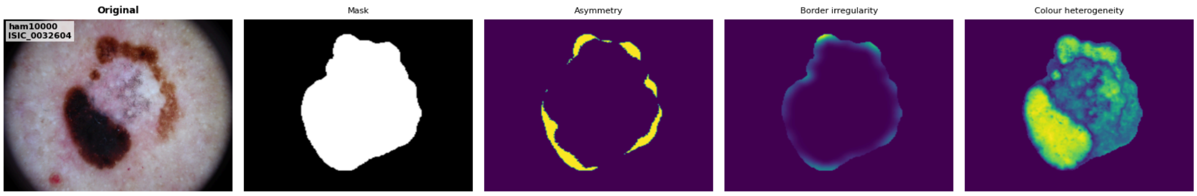

ham10000 / ISIC_0024767
Original
Mask
Grad-CAM
SHAP
LIME
ham10000
ISIC_0024767
ham10000 / ISIC_0032604
Original
Mask
Grad-CAM
SHAP
LIME
ham10000
ISIC_0032604
ham10000 / ISIC_0030276
Original
Mask
Grad-CAM
SHAP
LIME
ham10000
ISIC_0030276
ham10000 / ISIC_0026042
Original
Mask
Grad-CAM
SHAP
LIME
ham10000
ISIC_0026042
ham10000 / ISIC_0028162
Original
Mask
Grad-CAM
SHAP
LIME
ham10000
ISIC_0028162
isic2018 / ISIC_0013207
Original
Mask
Grad-CAM
SHAP
LIME
isic2018
ISIC_0013207
isic2018 / ISIC_0010058
Original
Mask
Grad-CAM
SHAP
LIME
isic2018
ISIC_0010058

ham10000 / ISIC_0031230

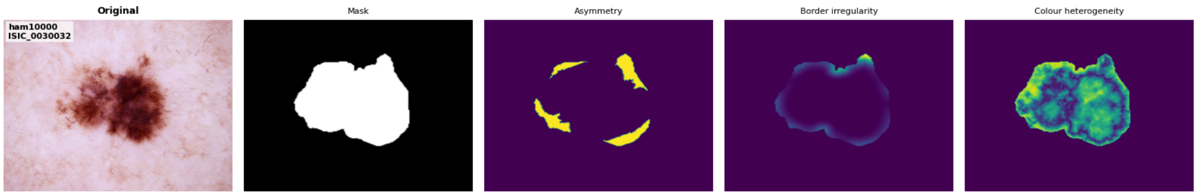


1.2 Pseudo-concepts

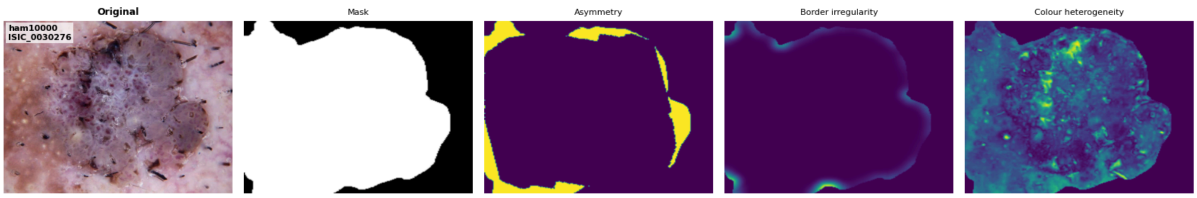
ham10000 / ISIC_0032604



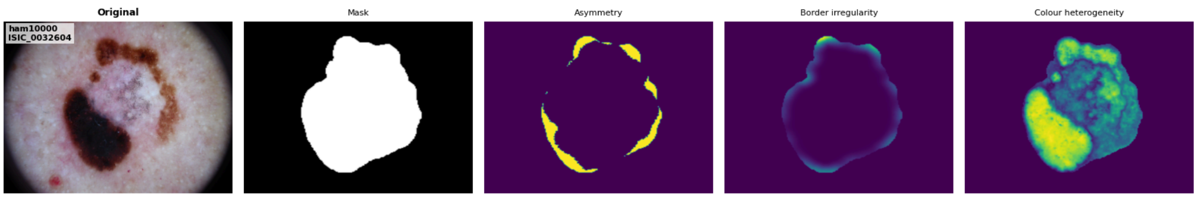
ham10000 / ISIC_0030032



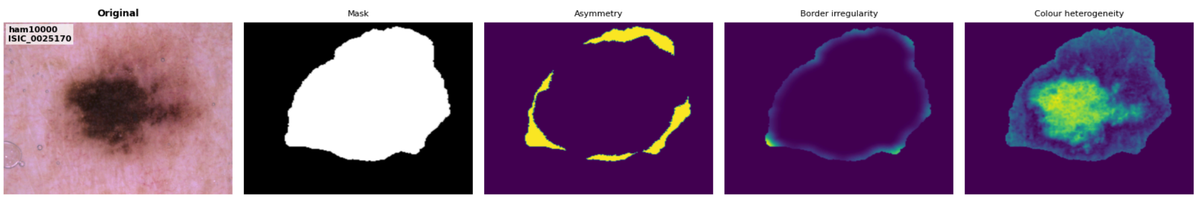
ham10000 / ISIC_0030276

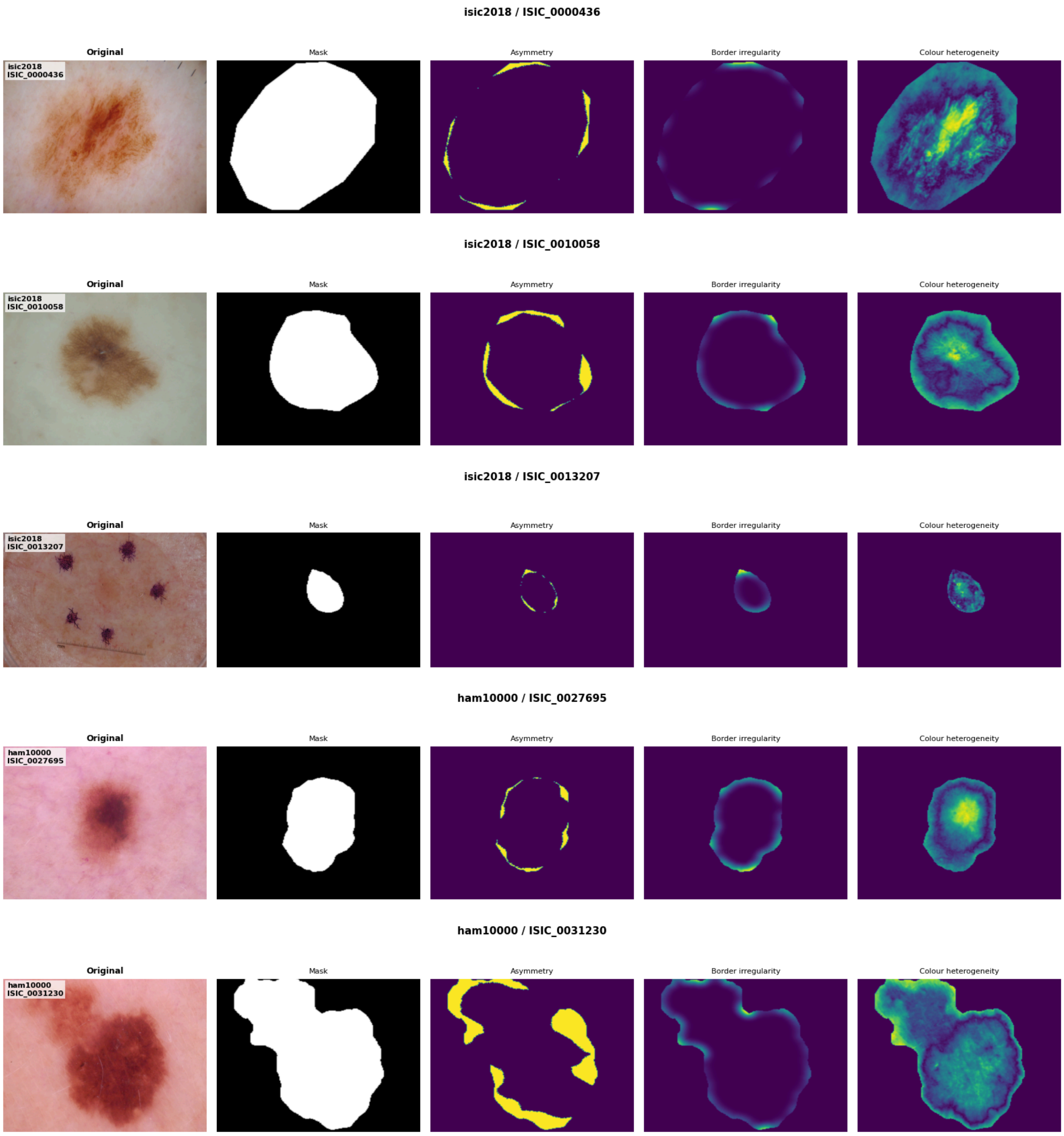


ham10000 / ISIC_0032604



ham10000 / ISIC_0025170




isic2018 / ISIC_0000436
Original
Mask
Asymmetry
Border irregularity
Colour heterogeneity
isic2018
ISIC_0000436
isic2018 / ISIC_0010058
Original
Mask
Asymmetry
Border irregularity
Colour heterogeneity
isic2018
ISIC_0010058
isic2018 / ISIC_0013207
Original
Mask
Asymmetry
Border irregularity
Colour heterogeneity
isic2018
ISIC_0013207
ham10000 / ISIC_0027695
Original
Mask
Asymmetry
Border irregularity
Colour heterogeneity
ham10000
ISIC_0027695
ham10000 / ISIC_0031230
Original
Mask
Asymmetry
Border irregularity
Colour heterogeneity
ham10000
ISIC_0031230

Section 2 - Metrics formulas

All summations are taken over the spatial pixel indices (i, j) of the saliency map and reference mask.

Name / Definition	Formula	Interpretation
H: Saliency Map: Normalised continuous saliency map generated by an XAI method.	$H = \{H_{i,j} \mid 0 \leq H_{i,j} \leq 1\}$	
M: Lesion mask: Binary reference mask representing either the lesion or a pseudo-concept region.	$M_{i,j} \in \{0, 1\}$	
X: Binary top-k% saliency region derived from the continuous saliency map H	$X_{i,j} \in \{0, 1\}$ where: $X_{i,j} = \mathbf{1}[H_{i,j} \geq \tau_k(H)]$ and: $\tau_k(H) = 1 - k$	
Intersection over Union; measures overlap between the binary top-k% XAI region and the reference mask relative to their union.	$IoU(X, M) = \dfrac{\sum_{i,j} X_{i,j} M_{i,j}}{\sum_{i,j}(X_{i,j} + M_{i,j} - X_{i,j} M_{i,j})}$	0 = no overlap 1 = identical binary regions
Dice: Measures similarity between the binary top-k% XAI region and the reference mask, giving greater weight to overlap.	$Dice(X, M) = \dfrac{2 \sum_{i,j} X_{i,j} M_{i,j}}{\sum_{i,j} X_{i,j} + \sum_{i,j} M_{i,j}}$	0 = no overlap 1 = identical binary regions
SIR (Saliency Inside Ratio); proportion of total continuous saliency mass located inside the reference region.	$SIR(H, M) = \dfrac{\sum_{i,j} H_{i,j} M_{i,j}}{\sum_{i,j} H_{i,j}}$	0 = no saliency inside 1 = all saliency mass inside
$TIxAI_{norm}(H, M))$: Normalised relevance score comparing mean continuous saliency inside and outside the reference region.	$TIxAI_{norm}(H, M) = \dfrac{RD_{In}}{RD_{In} + RD_{Out} + \epsilon}$	<0.5: outside dominant = 0.5: equal density > 0.5: inside dominant
RD_{In}: Mean continuous saliency within the reference region	$RD_{In} = \dfrac{\sum_{i,j} H_{i,j} M_{i,j}}{\sum_{i,j} M_{i,j}}$	
RD_{Out} : Mean continuous saliency outside the reference region	$RD_{Out} = \dfrac{\sum_{i,j} H_{i,j}(1 - M_{i,j})}{\sum_{i,j}(1 - M_{i,j})}$	

Section 3 - Summary of main results

Finding 1: SHAP achieves strongest lesion localisation.

Method	Mean IoU	Mean Dice	Mean SIR	Mean TIxAI
Grad-CAM	0.226	0.336	**0.427**	0.532
LIME	0.228	0.351	0.360	0.670
SHAP	**0.264**	**0.400**	0.394	**0.727**

SHAP achieves the highest lesion-level overlap scores. SHAP explanations may preserve finer local structure, better capture spatially meaningful regions, and produce more anatomically coherent saliency. This is consistent with SHAP's pixel attribution behaviour, compared to Grad-CAM's coarse activations.

Finding 2: Pseudo-concept alignment is much weaker than lesion localisation.

Concept	Best Method	Best Dice
Lesion Mask	**SHAP**	**0.400**
Asymmetry	SHAP	0.052
Border (best variant)	SHAP	0.025
Colour Heterogeneity	SHAP	0.118

Finding 3: Colour heterogeneity is the strongest pseudo-concept.

Method	Dice	TIxAI
Grad-CAM	0.083	0.397
LIME	0.080	0.574
SHAP	**0.118**	**0.668**

Colour heterogeneity is the pseudo-concept (but without ground-truth) most strongly aligned with XAI explanations.

Finding 4: Lesion size influences metrics more than diagnosis.

Lesion Size	Grad-CAM	LIME	SHAP
Normal	0.052	**0.062**	0.060
Small	0.017	0.015	**0.027**
Tiny	0.006	0.007	**0.011**
Very Tiny	0.004	0.004	**0.008**

Concept-level overlap decreases sharply as lesion size decreases.

Finding 5: Chance-adjusted evaluation provides a stricter assessment.
Dice Enrichment (Chance-Adjusted)

Method	Raw Asymmetry Dice	Dice Enrichment
Grad-CAM	0.025	0.82
LIME	0.027	**0.87**
SHAP	**0.028**	0.66

After correcting for random overlap, some apparent alignments disappear, while colour heterogeneity remains enriched above chance.